

Demonstration of an online digital twin for cost-effective operation of a water reuse pilot system

COPYRIGHT NOTICE

This manuscript has been authored by UT-Battelle, LLC, under contract DE-AC05-00OR22725 with the US Department of Energy (DOE). The US government retains and the publisher, by accepting the article for publication, acknowledges that the US government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so for US government purposes. DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (<http://energy.gov/downloads/doe-public-access-plan>).

This manuscript is intended for submission to ACS ES&T Engineering Journal as a special issue of Circular Water Economy.

This page is to be removed by editor upon acceptance.

Demonstration of an online digital twin for cost-effective operation of a water reuse pilot system

Subrata Mukherjee^{*1}, Pooria Ghorbani², Alex Melin¹, Sally Ghanem¹, Nader Rezaei³, Han Gu⁴, Megan Plumlee⁴, Brian Tarroja², Diego Rosso², and Kris Villez¹

¹Energy Science & Technology Directorate, Oak Ridge National Laboratory, Oak Ridge, TN

²Department of Civil and Environmental Engineering, University of California, Irvine, CA

³AECOM, Irvine, CA

⁴Orange County Water District, Fountain Valley, CA

*Email: mukherjees2@ornl.gov

Abstract

Modern water reuse systems face increasing operational complexity due to dynamic energy tariffs and tightening sustainability targets. This study presents the development and deployment of an online digital twin operating in continuous interaction with a membrane pilot system. The digital twin is based on an empirical model that is updated hourly with the latest sensor data. To this end, a sparse interpretable machine learning model is used to forecast power consumption as a function of permeate, concentrate, and recirculation flowrates. An optimization routine determines energy-efficient operating setpoints under process and safety constraints at every hour. A receding-horizon implementation enables adaptive operation under changing conditions. Over a two-week (approximately 332-hour) deployment, the framework achieved an average 38.33% reduction in specific energy cost relative to nominal steady-state operation, while maintaining stable actuator behavior. Temporal evolution of model parameters indicated convergence toward a compact and interpretable structure, demonstrating the robustness and stability of the digital twin during extended operation. These results establish the practical feasibility of digital twins based on empirical models for energy-aware, autonomous control in advanced water reuse systems.

Keywords

Cost minimization; Energy efficiency; Desalination; Bi-directional digital twin; Optimal production scheduling; Real-time closed-loop control; Adaptive L_1 -regularization

Highlights

- A digital twin was developed for adaptive online optimization of a membrane water reuse pilot system.
- An empirical modeling and receding-horizon control framework enables hourly adaptive operation.
- A two-week (332-hour) deployment achieved an average 38% reduction in specific energy cost.
- Stable coefficient evolution demonstrated robust and interpretable digital twin performance under dynamic conditions.

1 Introduction

Since the advent of water-quality sensor technologies in the 1980s, water production and reuse systems have become increasingly instrumented, generating large volumes of high-frequency operational data. Modern treatment facilities routinely collect thousands of sensor signals spanning flowrates, pressures, water quality indicators, and energy consumption. Historically, municipal wastewater treatment plants have leveraged these data to deploy advanced control strategies that address diurnal and seasonal variability in influent loading [1–3]. In contrast, water production and reuse systems have traditionally operated under near steady-state conditions, relying on buffer storage and relatively stable resource pricing to absorb operational variability. As a result, many facilities continue to operate at fixed setpoints despite growing external drivers for more flexible and adaptive operation.

Electricity price volatility has recently emerged as a significant operational constraint for advanced water reuse and desalination systems [1, 4–8]. Time-of-use tariffs introduce strong temporal variation in operating costs, yet most treatment trains remain insensitive to these signals, operating with static flowrate schedules that fail to exploit periods of lower electricity prices [9–11]. Previous studies have demonstrated the potential benefits of energy-aware operation in reverse osmosis systems, including improved integration with renewable energy sources and reduced electricity expenditure [12, 13]. However, practical implementation of closed-loop control strategies in water reuse systems remains limited, and sustained online demonstrations under operational constraints are rare. Importantly, most water reuse systems are equipped with upstream and downstream buffer storage, enabling flexible operation without compromising water supply requirements. Within this flexibility, operational variables such as permeate, concentrate, and recirculation flowrates directly influence power demand in reverse osmosis systems, providing practical control levers for reducing electricity costs under dynamic tariffs.

Digital twins have emerged as a promising paradigm for enabling adaptive and data-driven operation of complex infrastructure systems. A digital twin is typically defined as a virtual representation of a physical asset that is continuously updated through data exchange with its physical counterpart, enabling monitoring, prediction, optimization, and control [1, 2, 14–17]. Continuous updating is particularly important in water reuse systems, where internal process dynamics such as membrane fouling, equipment aging, and nonlinear hydraulic behavior evolve over time. Unlike digital shadows, which support only one-way data flow from the physical system to the virtual model, digital twins require bidirectional interaction that enables recommended control actions to be transferred back to the physical system [18–20]. Such bidirectional coupling is essential for predictive control and adaptive optimization in engineered systems exposed to continuously changing operating conditions [21, 22].

Within the water technologies sector, digital twins have been applied across a range of domains including wastewater treatment, drinking water systems, pump scheduling, desalination, and water distribution management [23–26]. Although these studies demonstrate the potential of digital twins for monitoring and decision support, many reported implementations remain offline, unidirectional, or simulation-based. As a result, there remains a gap between conceptual digital twin architectures and practical deployments [1] capable of sustained autonomous operation under real-world operational constraints.

To address this gap, this work develops and deploys a bidirectional digital twin framework based on an empirical model-based optimization (EMBO) pipeline [27, 28]. The framework employs adaptive sparse regression to capture the relationship between controllable flowrates and power consumption, and integrates this model within a receding-horizon optimization scheme that accounts for time-varying electricity tariffs. The approach is validated through a continuous two-week deployment on a pilot system replicating the operational characteristics of the Orange County Water District (OCWD) advanced water reuse facility. By demonstrating reliable real-time operation, sustained cost reduction, and stable closed-loop behavior using conventional computational hardware, this study advances digital twin technology from conceptual promise toward practical, scalable deployment for autonomous, energy-aware water reuse operations.

2 Materials and Methods

This section describes the three core components of the proposed digital twin framework and the methods used to evaluate its performance: (i) the pilot system serving as the physical counterpart of the full-scale plant, (ii) a data-driven empirical modeling and optimization pipeline forming the digital twin, and (iii) the

closed-loop feedback mechanism that couples the physical and virtual domains. Together, these components enable online adaptive optimization of system operation under time-varying electricity tariffs.

The digital twin operates according to a model predictive control (MPC) paradigm characterized by two principal horizons. An estimation horizon consisting of the most recent seven days of operational data is used for adaptive model calibration and parameter updating. A prediction or control horizon of 168 hours (one week) is used during optimization to evaluate tariff-weighted operating cost under anticipated conditions. At each hourly optimization step, the empirical model is retrained using the updated estimation horizon and a new 168-hour operating trajectory is computed. Only the first-hour control action from this optimized trajectory is implemented on the physical pilot, after which the optimization procedure is repeated using newly acquired measurements. This receding-horizon implementation enables continuous adaptation to evolving operating conditions and electricity tariffs.

Figure 1 illustrates the overall system architecture. Operational data collected from the pilot are stored in a cloud-based environment and supplied to the digital twin at hourly intervals, with each training window defined as the most recent seven days of operation. The pipeline couples an empirical model for power prediction with a cost-aware optimization module that computes flowrate setpoints over the 168-hour prediction horizon, subject to process and safety constraints.

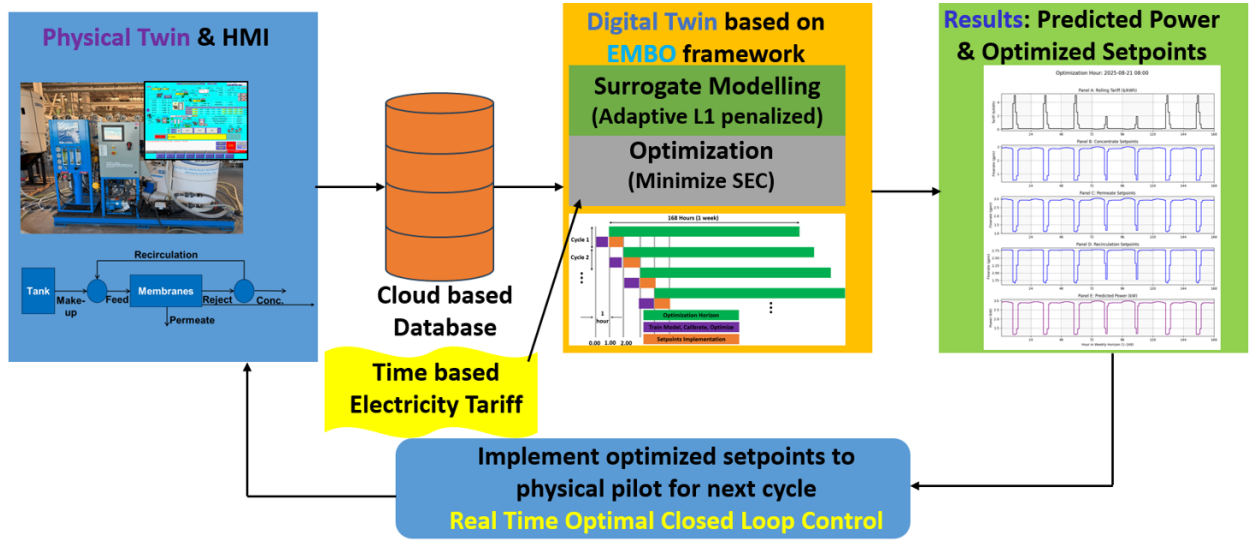


Figure 1: Bi-directional digital twin framework for online energy-aware operation of a water reuse pilot. The pipeline integrates hourly empirical model calibration and cost-aware optimization over a 168-hour prediction horizon, with first-hour setpoints implemented on the physical system in a receding-horizon manner.

2.1 Physical Pilot System and Measurements

Figure 2(a) illustrates the physical pilot system used in this study, which mimics the operational characteristics of the full-scale Orange County Water District (OCWD) advanced water reuse facility. The pilot consists of three treatment units arranged in series: (i) UF, (ii) RO, and (iii) ultraviolet and advanced oxidation processes (UV/AOP). The present work focuses on operational control within the RO unit, which is the dominant contributor to energy consumption and provides multiple controllable degrees of freedom.

Within the RO subsystem, three flowrate variables are manipulated: permeate (F_{perm}), concentrate (brine) (F_{conc}), and recirculation (F_{rctc}) flowrates (Figure 2(b)). These three flows determine all remaining internal flows, including membrane feed and reject streams, and therefore provide a minimal set of decision variables for energy-aware operation. The measurements used by the digital twin include online observations of F_{perm} , F_{conc} , and F_{rctc} , together with the aggregate electrical power demand of the pilot system, which serves as the measured response variable.

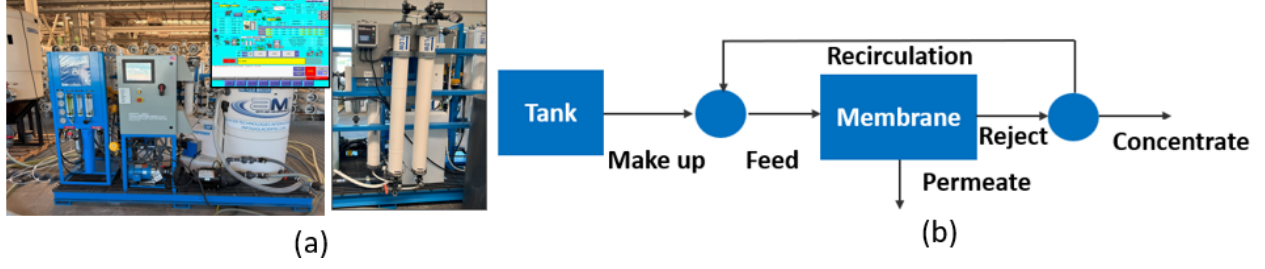


Figure 2: (a) Physical pilot system comprising UF, RO, and UV/AOP units. (b) Schematic flow configuration of the RO subsystem highlighting the permeate, concentrate, and recirculation flowrate decision variables.

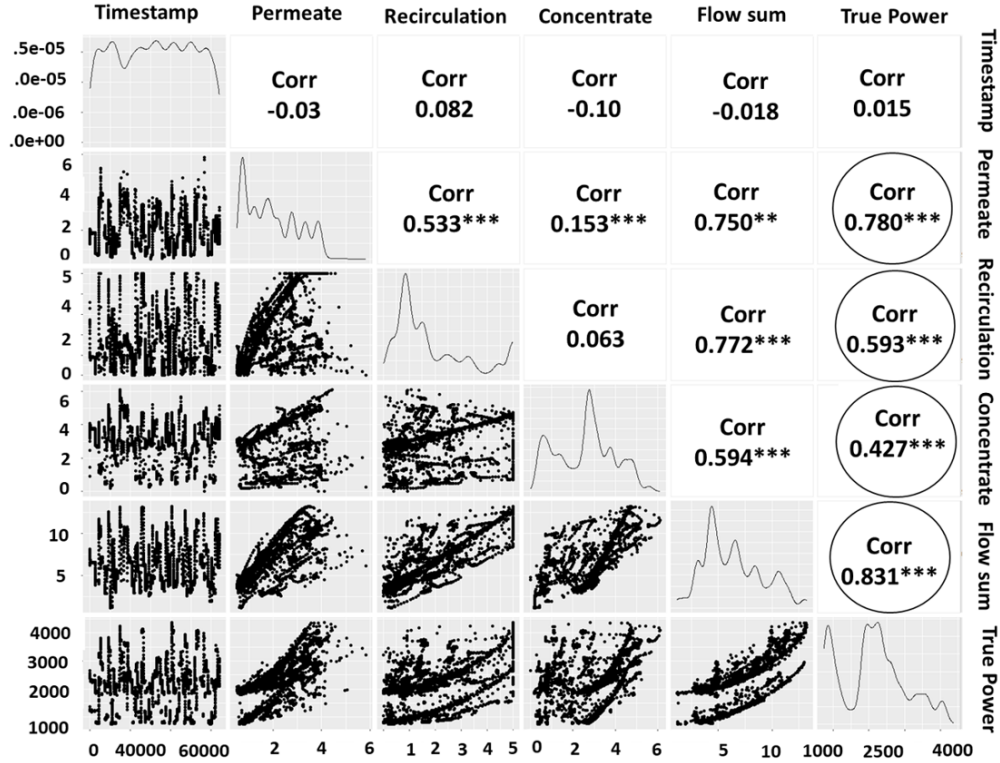


Figure 3: Pairwise scatter-plot matrix illustrating empirical relationships between aggregate power demand and flowrate variables. Pearson correlation coefficients demonstrate positive associations between power consumption and permeate, concentrate, and recirculation flowrates.

2.2 Empirical Power Model and Exploratory Analysis

Figure 3 presents a model-free exploratory analysis of the measured data using a pairwise scatter-plot matrix. The analysis reveals a strong positive association between aggregate power demand and the total flowrate (Pearson correlation ≈ 0.83), as well as with permeate flowrate, whereas recirculation and concentrate flowrates exhibit moderate positive correlations. Based on inspection of these relationships, a nonlinear empirical model is developed to predict power draw from the flowrate signals. The observed structure of the data suggests that the flowrates provide informative predictors of system power consumption, and validation results indicate that this formulation yields reliable forecasting performance over seven-day training windows.

2.2.1 Regression Model Construction

Let $\mathbf{X} \in \mathbb{R}^{n \times 3}$ denote the input matrix comprising permeate, concentrate, and recirculation flowrates, and let $Y \in \mathbb{R}^n$ denote the corresponding measured aggregate power demand. To capture nonlinearities and cross-coupling between decision variables, we construct a polynomial feature expansion up to cubic order ($d = 3$),

$$\Phi(\mathbf{X}) = [X_1, X_2, X_3, X_1^2, X_1X_2, X_1X_3, \dots], \quad (1)$$

yielding a library of 19 candidate basis functions. Because this representation can overfit when training data are limited (especially during early operation), model identification is performed using L_1 -regularized regression via the Least Absolute Shrinkage and Selection Operator (LASSO) [29–33], which promotes sparsity and interpretability. Specifically, regression coefficients β are estimated by:

$$\hat{\beta} = \arg \min_{\beta} \{ \|\mathbf{Y} - \Phi(\mathbf{X})\beta\|_2^2 + \lambda \|\beta\|_1 \}, \quad (2)$$

where $\lambda > 0$ controls the sparsity-accuracy trade-off.

2.2.2 Structured λ Selection with Degeneracy Safeguards

Model training uses 10-fold cross-validation [34–37] over a logarithmically spaced grid $\lambda \in [10^2, 10^{-4}]$. Let $\lambda_{\min}$ denote the regularization parameter that minimizes the cross-validated mean squared error (CV MSE), and let λ_{1se} denote the largest λ whose CV MSE lies within one standard error of the minimum. In early deployment, limited training data can yield unstable or overly sparse models (including degenerate solutions with all coefficients shrunk to zero). To ensure consistent model availability, EMBO employs a structured three-tier selection rule complemented by two degeneracy safeguards:

1. **Primary rule:** select $\lambda_{\min}$ if the number of non-zero coefficients at $\lambda_{\min}$ is ≤ 8 .
2. **Secondary rule:** otherwise scan from $\lambda_{\min}$ toward λ_{1se} and select the first λ yielding ≤ 7 non-zero coefficients.
3. **Fallback rule:** if no such λ exists, set $\lambda_{\text{opt}} = \lambda_{1se}$.
4. **Safeguard 1:** if $\lambda_{\text{opt}} = \lambda_{1se}$ yields a fully sparse model, revert to $\lambda_{\min}$.
5. **Safeguard 2:** if $\lambda_{\min}$ also yields zero coefficients, a minimal linear model is constructed via forward stepwise selection on the original (non-polynomial) inputs to prevent control degeneracy.

The complete decision logic is summarized in Figure 4.

2.3 Empirical Feasibility Constraints for Safe Setpoint Generation

For safe closed-loop operation, it is necessary to define an operating region that is both experimentally realizable and consistent with equipment limitations. An initial experimental campaign generates candidate setpoints using Latin hypercube sampling within the operational bounds specified in the pilot manual. During inspection of the recorded data, several setpoint combinations, while satisfying prescribed safety limits, cannot be physically realized. In particular, the single pump in the RO unit is unable to consistently deliver certain requested membrane feed flowrates.

To account for this limitation, we identify the subset of experimentally feasible operating points and derive linear inequality constraints describing their convex envelope. These feasibility constraints are incorporated into the downstream optimization problem to prevent the digital twin from proposing operationally unattainable setpoints.

To quantify realizability, we segment the experimental time series into steady-state blocks in which the setpoint flowrates ($F_{\text{perm,set}}, F_{\text{conc,set}}, F_{\text{rcyc,set}}$) remain constant. Within each block, we compute median measured flowrates ($F_{\text{perm,med}}, F_{\text{conc,med}}, F_{\text{rcyc,med}}$) to reduce sensitivity to transients and measurement noise. To measure the closeness of the observed flow rates to the requested flow rate setpoint combination, the

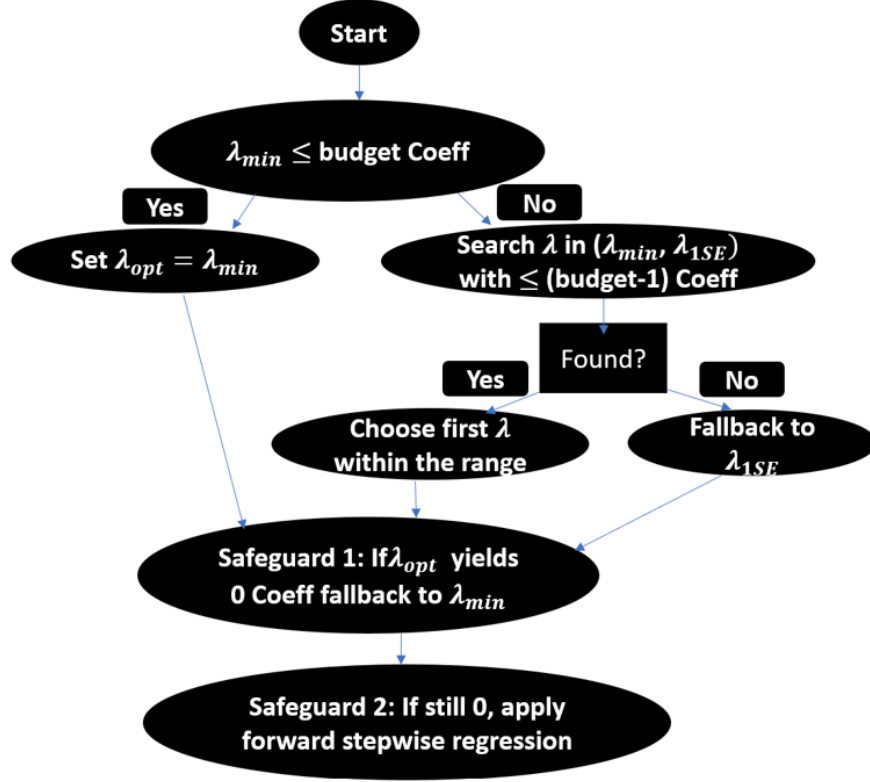


Figure 4: Schematic illustrating the adaptive L_1 -regularization strategy and structured fallback logic used for empirical model identification.

maximum absolute deviation (an ℓ_∞ metric) is defined as $d_{\max} = \max \left\{ |F_{\text{perm, set}} - F_{\text{perm, med}}|, |F_{\text{conc, set}} - F_{\text{conc, med}}|, |F_{\text{rcyc, set}} - F_{\text{rcyc, med}}| \right\}$.

Smaller $d_{\max}$ indicates high setpoint fidelity, whereas larger values suggest infeasibility due to actuator limitations or coupling effects. Figure 5(a) visualizes all setpoints in 3D, colored by $d_{\max}$. Based on the empirical distribution of deviations, the feasible region is defined as $\mathcal{F} = \{\mathbf{F}_{\text{set}} \in \mathbb{R}^3 \mid d_{\max} < 0.2\}$, where $\mathbf{F}_{\text{set}} = (F_{\text{perm, set}}, F_{\text{conc, set}}, F_{\text{rcyc, set}})^\top$ and the 0.2 gpm threshold reflects a conservative tolerance under which tracking is consistently reliable.

To obtain a compact representation suitable for constrained optimization, the 3D convex hull of feasible setpoints is computed using standard computational geometry algorithms. Each hull facet defines a supporting half-space

$$a_{\text{perm}} F_{\text{perm}} + a_{\text{conc}} F_{\text{conc}} + a_{\text{rcyc}} F_{\text{rcyc}} \leq b, \quad (3)$$

and the collection of these inequalities defines a polyhedral approximation of the empirically realizable operating region. The complete list of hull-derived inequality coefficients is provided in the Section A (Table 4).

Figure 5(b) shows the 3D setpoints colored by whether they satisfy all hull-derived half-spaces. Points satisfying all convex-hull inequalities are marked in green, while violations appear in red. For additional physical interpretability, we project setpoints into aggregate flow coordinates $F_{\text{feed}} = F_{\text{perm}} + F_{\text{conc}} + F_{\text{rcyc}}$ and $F_{\text{reject}} = F_{\text{conc}} + F_{\text{rcyc}}$, and visualize the corresponding 2D feasibility envelope (Figure 5(c)). Finally, Figure 5(d) depicts the polyhedral convex hull itself, highlighting boundary points (in purple) that generate the hull facets and thus directly correspond to the linear constraints used during optimization.

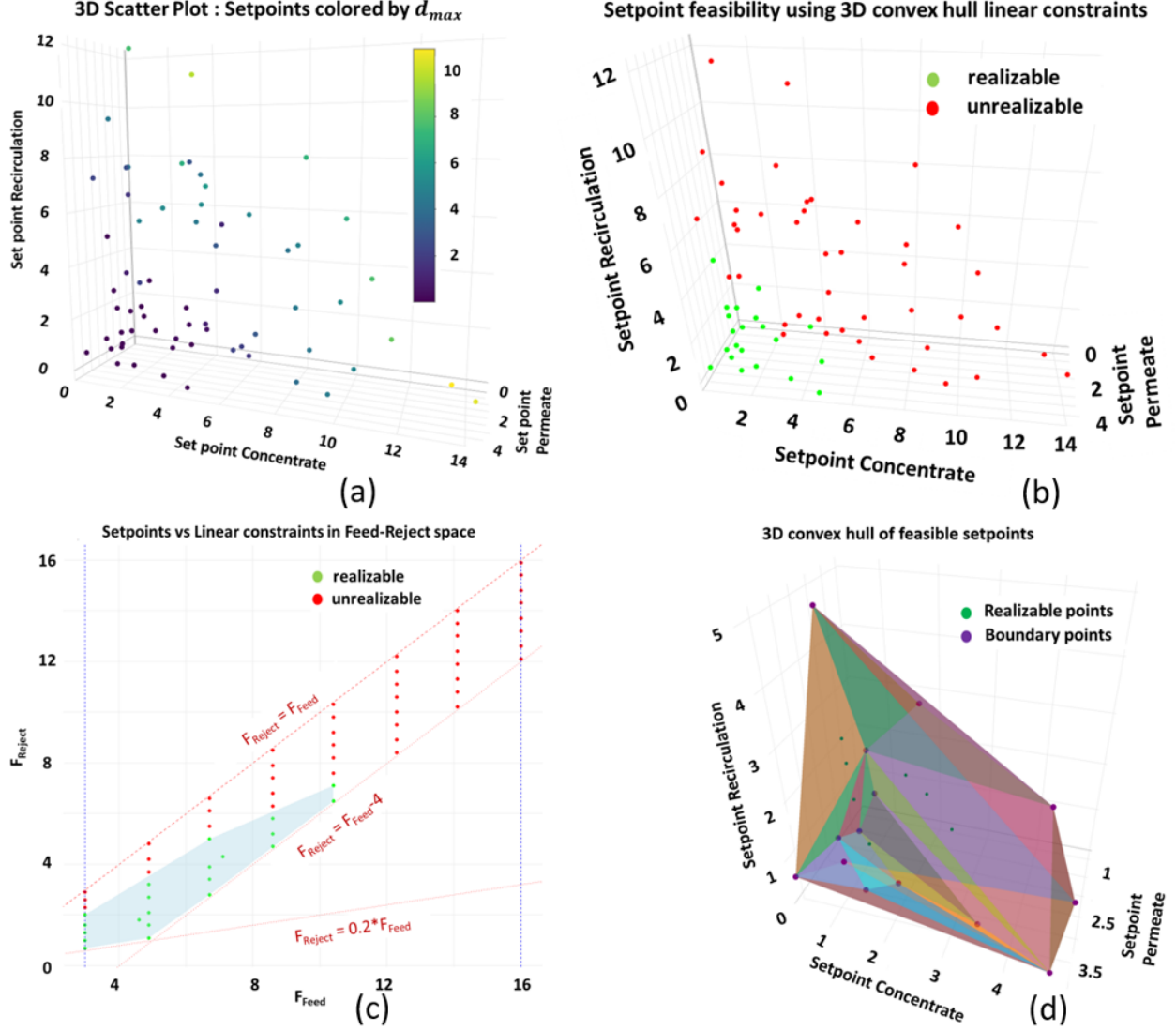


Figure 5: (a) 3D view of setpoint realizability colored by $d_{\max}$. (b) Setpoints colored by feasibility under hull-derived linear constraints. (c) Projection into feed–reject space. (d) Convex polytope and boundary points defining the hull facets.

2.4 Digital Twin for Online Optimization

At each hourly control cycle, the digital twin updates the empirical power model using the most recent estimation horizon and computes a 168-hour flowrate schedule that minimizes tariff-weighted specific energy cost while meeting production targets and operational constraints. Only the first-hour setpoints are implemented on the pilot via the programmable logic controller (PLC), after which new measurements are acquired and the procedure repeats, yielding receding-horizon operation.

2.4.1 Power Forecasting and Cost Evaluation

Given a candidate flowrate trajectory $\mathbf{X}_{\text{cand}} \in \mathbb{R}^{168 \times 3}$ over the 168-hour prediction horizon, predicted power demand is computed as $\hat{\mathbf{Y}} = \Phi(\mathbf{X}_{\text{cand}})\hat{\boldsymbol{\beta}}$. The sparse structure of $\hat{\boldsymbol{\beta}}$ enables interpretability by identifying

dominant flowrate terms and interactions.

$$J(\mathbf{X}_{\text{cand}}) = \sum_{t=1}^{168} \hat{Y}_t c_t.$$

Given a candidate flowrate trajectory $\mathbf{X}_{\text{cand}} \in \mathbb{R}^{168 \times 3}$ over the 168-hour planning horizon, predicted power demand is computed as $\hat{\mathbf{Y}} = \Phi(\mathbf{X}_{\text{cand}})\hat{\beta}$. The sparse structure of $\hat{\beta}$ enables interpretability by identifying dominant flowrate terms and interactions. Figure 6 illustrates feature importance derived from the model after eight hours of operation, where only 10 of the 19 candidate terms remain active (approximately 47% sparsity). First-order flowrate terms dominate the learned mapping, indicating physically consistent sensitivity of power demand to operational setpoints.

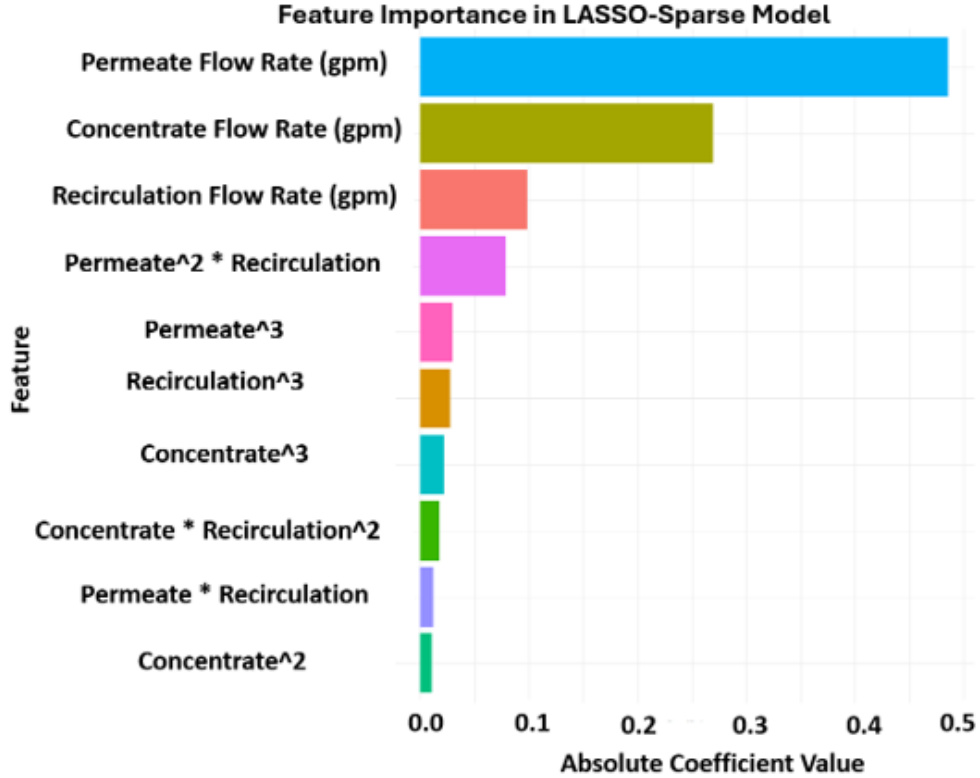


Figure 6: Feature importance derived from the sparse empirical model after 8 hours of operation.

2.4.2 Optimization Formulation

Tariff-weighted cost is evaluated using hourly electricity tariffs $\mathbf{c} \in \mathbb{R}^{168}$. The decision variable is the hourly flowrate trajectory $\mathbf{X}_{\text{cand}} = \{F_{\text{perm}}^{(t)}, F_{\text{conc}}^{(t)}, F_{\text{recyc}}^{(t)}\}_{t=1}^{168}$, yielding a $24 \times 7 \times 3 = 504$ -dimensional optimization problem. The optimization is posed as

$$\min_{\mathbf{X}_{\text{cand}} \in \mathbb{R}^{168 \times 3}} \sum_{t=1}^{168} \left(\Phi(\mathbf{X}_{\text{cand}}^{(t)})\hat{\beta} \right) c_t, \quad (4)$$

subject to equipment and operational constraints, including:

$$0 \leq F_{\text{perm}}^{(t)} \leq 4, \quad (5)$$

$$3 \leq F_{\text{perm}}^{(t)} + F_{\text{conc}}^{(t)} + F_{\text{rcyc}}^{(t)} \leq 16, \quad (6)$$

$$F_{\text{perm}}^{(t)} \leq 4 \left(F_{\text{conc}}^{(t)} + F_{\text{rcyc}}^{(t)} \right), \quad (7)$$

$$(1 - R_{\text{target,h}}) F_{\text{perm}}^{(t)} - R_{\text{target,h}} F_{\text{conc}}^{(t)} - R_{\text{target,h}} F_{\text{rcyc}}^{(t)} \leq 0, \quad \forall t, \quad (8)$$

$$\sum_{t=1}^{168} F_{\text{perm}}^{(t)} \Delta t = V_{\text{target}}, \quad (9)$$

$$(1 - R_{\text{target}}) \sum_{t=1}^{168} F_{\text{perm}}^{(t)} = R_{\text{target}} \sum_{t=1}^{168} F_{\text{conc}}^{(t)}. \quad (10)$$

Here $R_{\text{target,h}}$ denotes the hourly recovery target, R_{target} the weekly average recovery target, and V_{target} the required cumulative permeate volume.

The nonlinear constrained optimization is solved using the trust-region constrained method (**trust-constr**) implemented in SciPy [38–42]. The overall receding-horizon control architecture is illustrated in Figure 7 [43].

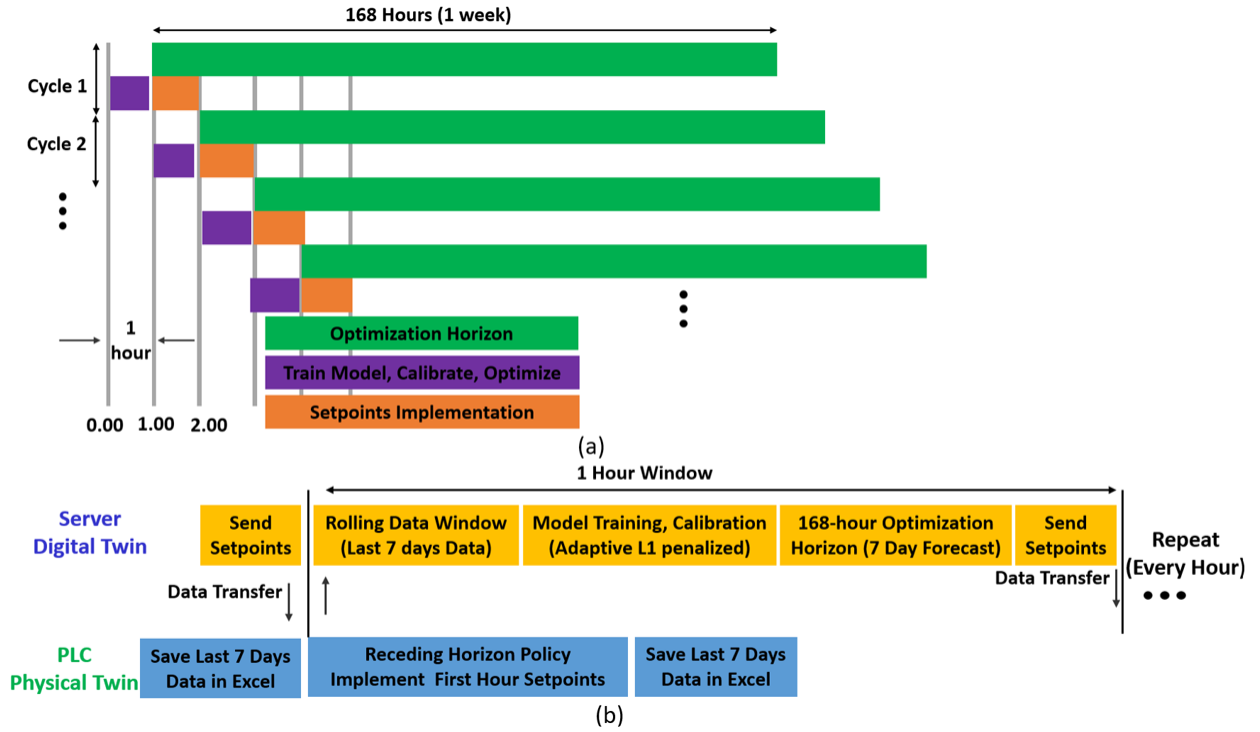


Figure 7: Digital twin control architecture: (a) overall closed-loop framework, (b) hourly receding-horizon control cycle.

2.4.3 Post-Optimization Perturbation of Flowrate Schedules

In early operation or under weak tariff gradients, the optimizer may return nearly constant flowrate trajectories across the prediction horizon. If such trajectories are repeatedly implemented, the applied setpoints remain effectively constant over a complete estimation horizon, resulting in an uninformative dataset for

subsequent model identification. To mitigate this issue while preserving operational feasibility, EMBO applies a bounded random perturbation to the optimized schedule. Let $\mathbf{Q}_{\text{opt}} \in \mathbb{R}^{168 \times 3}$ denote the optimized flowrate schedule and let $\mathbf{Q}_{\text{min}}, \mathbf{Q}_{\text{max}} \in \mathbb{R}^3$ denote the corresponding physical bounds. For each flow channel $j \in \{1, 2, 3\}$, the perturbation range is defined as $\Delta Q_j = (Q_{\text{max}}^{(j)} - Q_{\text{min}}^{(j)}) sf$, where s is a shrinkage fraction and f is a perturbation fraction. In the present implementation, the values $s = 0.05$ and $f = 0.98$ are used. A perturbation matrix $\mathbf{P} \in \mathbb{R}^{168 \times 3}$ is then sampled element-wise as $P_{tj} \sim \mathcal{U}(-\Delta Q_j, \Delta Q_j)$, and the perturbed schedule is obtained as $\mathbf{Q}_{\text{pert}} = \mathbf{Q}_{\text{opt}} + \mathbf{P}$.

After perturbation, setpoints are projected back onto the feasible region if necessary and rounded to enforce actuator-friendly updates prior to implementation.

3 Results and Discussion

This section presents and discusses the results obtained from the two week continuous experimental deployment of the proposed digital physical twin framework, corresponding to approximately 332 uninterrupted hours of real time operation. The analysis focuses on the economic performance, operational behavior, and computational feasibility of the optimization under dynamic electricity tariffs.

The behavior of the structured λ selection rule during early deployment is illustrated in Figure 8. The figure shows that at eight hours of operation, cross-validation error bands tighten and conventional choices such as λ_{lse} yield stable sparse models (Figure 8a–b). In contrast, after only one hour of operation, large uncertainty in cross-validation estimates can render λ_{lse} overly conservative, leading to very few active terms (Figure 6c–d).

Table 1 summarizes representative values of λ_{min} , λ_{lse} , and the selected λ_{opt} , together with corresponding cross-validation mean square errors and sparsity levels. These results demonstrate that the structured selection rule prevents premature model collapse under data scarcity while preserving sparsity as training data accumulate.

Table 1: Representative cross-validation outcomes illustrating structured λ selection under limited (1 hour) and expanded (8 hours) training data.

Case	λ_{min}	λ_{lse}	λ_{opt}	MSE λ_{min}	MSE λ_{opt}	Nonzeros
8 hours	0.00009	0.0199	0.0199	0.2651	0.2722	10
1 hour	0.000125893	0.63095	0.00794	0.5445	0.8447	7

Figure 9 illustrates a representative optimization outcome selected at the 50th hour of deployment (approximately two days after initialization). This snapshot corresponds to a single 168 hour optimization horizon generated using the most recent model. As shown in Figure 9(a), the hourly electricity cost under optimized control is substantially reduced relative to nominal steady state operation, particularly during periods of elevated tariff. Importantly, the optimized flowrate schedules satisfy the same cumulative permeate production requirement over the 7-day horizon as the nominal operating policy, ensuring a fair comparison. Quantitatively, the optimized operating policy achieves a cost reduction of 41.39% relative to nominal operation for this representative horizon, as summarized in Table 2. The reduction reflects a decrease in specific energy cost driven by strategic load shifting while producing the same amount of water during the forecast horizon.

Table 2: Cost comparison between nominal steady-state operation and optimized control for the representative 168-hour horizon shown in Figure 9.

Default Cost (\$)	Optimized Cost (\$)	Cost Savings (\$)	Cost Reduction (%)
198.15	116.14	82.01	41.39

Additional optimization examples corresponding to the 53rd and 300th hours of operation are provided in Appendix A (Figures 16 and 17), illustrating the repeatability and robustness of the proposed framework

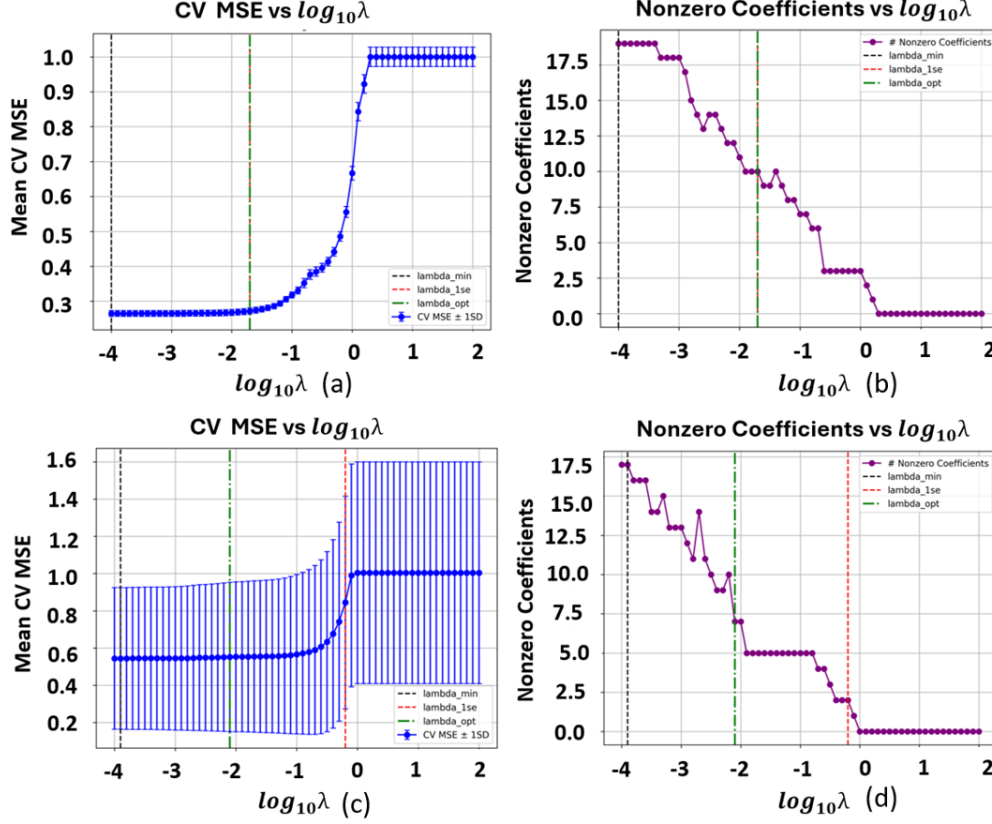


Figure 8: Cross-validation performance and sparsity behavior of the empirical model under different regularization levels: (a,b) results after 8 hours of operation; (c,d) results after 1 hour of operation.

across the full deployment period.

Figure 9(b) demonstrates how the predicted power demand adapts to the time varying tariff structure. The model driven controller systematically suppresses power consumption during high tariff periods while reallocating operational load to lower tariff intervals, confirming the intended economic responsiveness of the digital twin. This behavior is further reflected in Figures 9(c)–(e), which show the optimized 168-hour flowrate trajectories for concentrate, permeate, and recirculation streams. In contrast to the constant steady state setpoints used under nominal operation (shown in gray), the optimized flowrates dynamically decrease during high tariff intervals and increase during low tariff periods, consistent with cost aware operation across all manipulated variables.

Figure 9(f) illustrates the convergence behavior of the optimization routine for the same horizon. The objective function decreases monotonically and converges within a small number of iterations, typically requiring 5–10 minutes of computation on conventional hardware. This confirms that the full calibration optimization deployment cycle comfortably fits within the one hour control interval, enabling reliable real time operation.

Beyond this representative example, the cumulative performance over the full experimental campaign is summarized in Table 3. Across the entire 332-hour deployment, the proposed framework achieves an average cost reduction of 38.33% relative to nominal steady-state operation. The consistency of these savings over an extended period demonstrates the robustness and repeatability of the digital physical twin under sustained real time conditions.

Figure 10 illustrates the hourly flowrate setpoints implemented on the physical pilot throughout the full experimental deployment. At each control cycle, the digital twin recalibrates the empirical model using the most recent data and computes an updated 168-hour optimization horizon, of which only the first-hour setpoints are enacted on the pilot. The figure displays the applied optimized setpoints for concentrate (solid

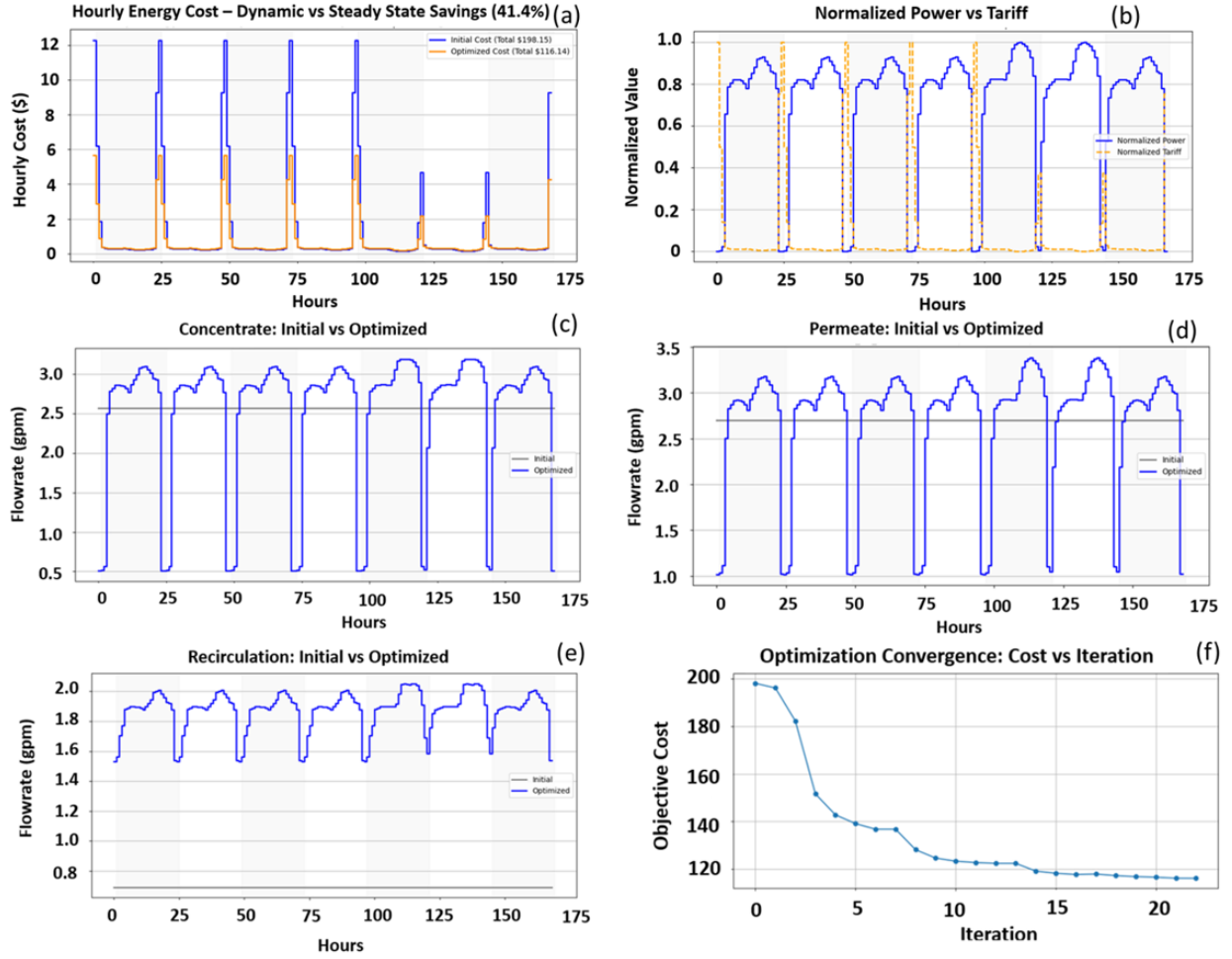


Figure 9: Representative optimization results at the 50th hour of deployment. (a) Hourly electricity cost under nominal steady-state operation (blue) and optimized control (orange), showing a cost reduction of 41.39%. (b) Predicted power demand (blue) overlaid with the normalized electricity tariff (orange), illustrating tariff-aware load shifting. (c) Concentrate flowrate profile under optimized control (blue) compared with steady-state operation (gray). (d) Permeate flowrate profile under optimized control (blue) compared with steady-state operation (gray). (e) Recirculation flowrate profile under optimized control (blue) compared with steady-state operation (gray). (f) Optimization convergence showing objective cost versus iteration number, demonstrating rapid convergence within the real-time control interval.

Table 3: Aggregate cost performance over the full 332-hour deployment.

Default Cost (K \$)	Optimized Cost (K \$)	Cost Reduction (%)
67	41	38.3

blue), permeate (solid green), and recirculation (solid red), together with the corresponding nominal steady-state flowrates (dashed lines). The results show a clear and consistent relationship between the applied setpoints and the time-varying electricity tariff. Across all three manipulated variables, the optimized setpoints are systematically reduced during high-tariff periods and increased during low-tariff periods, whereas the steady-state operation remains constant and unresponsive to tariff fluctuations.

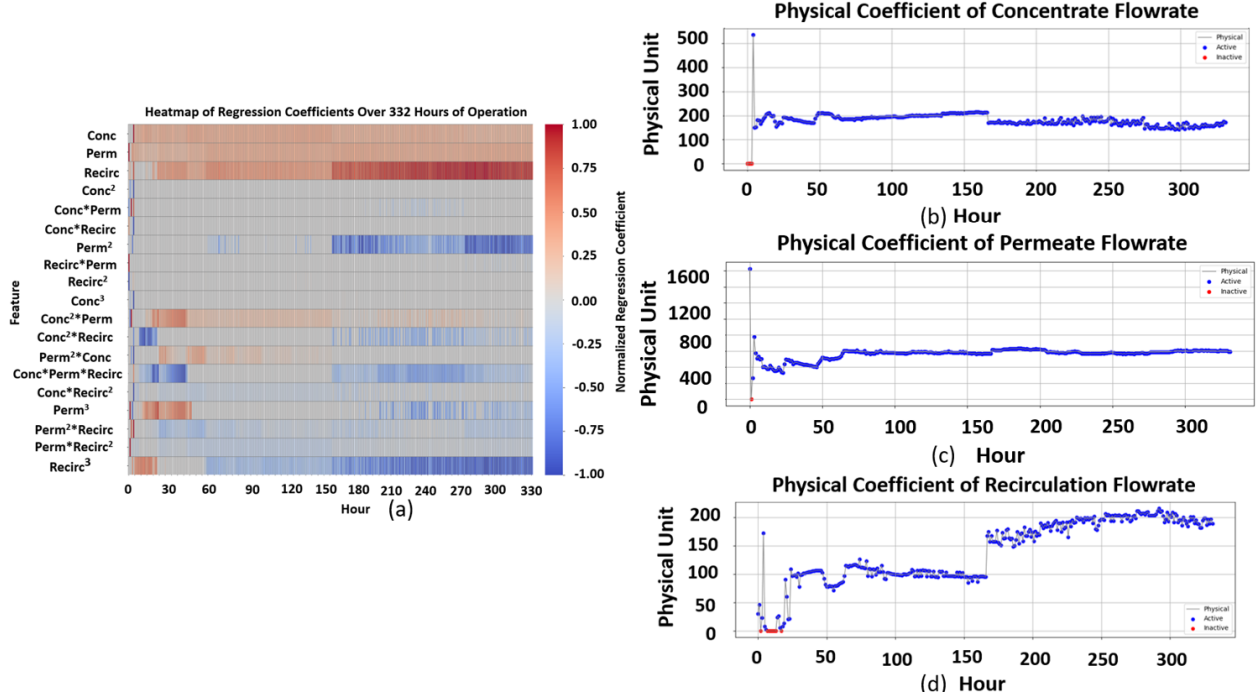


Figure 10: Temporal evolution of regression coefficients over 332 hours of real-time operation. (a) Heatmap of normalized physical regression coefficients, with each feature scaled to the interval $[-1, 1]$ based on its maximum absolute value across time. (b) Physical coefficient of the concentrate flowrate as a function of hour. (c) Physical coefficient of the permeate flowrate as a function of hour. (d) Physical coefficient of the recirculation flowrate as a function of hour. Active coefficients are shown in blue, while coefficients pruned by LASSO regularization are shown in red, highlighting structural sparsity and convergence of the model over time.

To evaluate the interpretability and temporal stability of the digital twin, we tracked the evolution of regression coefficients over the full 332-hour experimental deployment. At each hourly control cycle, the trained LASSO model was archived, enabling post hoc analysis of coefficient trajectories and structural sparsity over time. This analysis provides insight into the convergence behavior of the model as training data accumulate under real-time operation.

Because model training was performed on standardized data, the estimated regression coefficients $\hat{\beta}_j$ are expressed in normalized units. To recover physically interpretable coefficients β_j^{phys} (e.g., in units of W/gpm or W/gpm²), the coefficients are rescaled by reversing the affine transformations applied during standardization. Specifically, the physical coefficients are obtained as $\beta_j^{\text{phys}} = (\sigma_y / \sigma_{x_j}) \hat{\beta}_j$, where σ_y denotes the standard deviation of the measured power signal and σ_{x_j} denotes the standard deviation associated with the corresponding feature after polynomial expansion. This transformation maps the coefficients from normalized units back to their physical units.

Figure 11(a) summarizes the temporal evolution of all regression coefficients using a normalized heatmap representation. Each coefficient time series is normalized by its maximum absolute magnitude over the 332 hour window, $\hat{\beta}_j^{\text{phys}}(h) = \beta_j^{\text{phys}}(h) / \max_h |\beta_j^{\text{phys}}(h)|$, thereby rescaling all features to the interval $[-1, 1]$ for visual comparability. This representation highlights both coefficient activation patterns and relative temporal

variability across features.

Figures 11(b)–(d) further visualize the physical coefficients associated with the first-order flowrate terms: concentrate, permeate, and recirculation as explicit time series. Coefficients that were pruned by the LASSO regularization at a given hour are assigned a value of zero and shown in red, while active coefficients are shown in blue. This visualization directly illustrates the evolving structure under regular retraining.

Several consistent trends emerge from this analysis. First order flowrate coefficients remain persistently active throughout the deployment and exhibit relatively stable magnitudes after couple of hours, indicating convergence toward a reliable and physically meaningful model structure. In contrast, higher-order and interaction terms are activated intermittently, reflecting transient structural exploration during early learning stages followed by progressive sparsity enforcement as data volume increases. Certain higher order features appear only during specific temporal windows, suggesting localized relevance under transient operating conditions or tariff regimes. In later stages of the experiment, the number of active coefficients decreases and coefficient variability diminishes, indicating stabilization of model complexity and improved robustness of the learned empirical model.

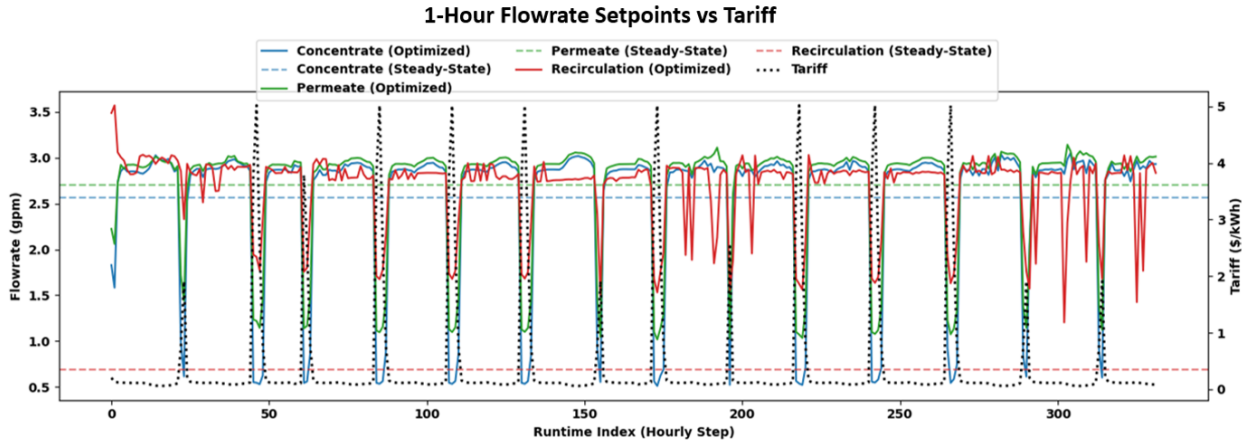


Figure 11: Hourly flowrate setpoints applied to the physical pilot over the full experimental deployment. Solid lines denote optimized setpoints implemented at each hour for concentrate (blue), permeate (green), and recirculation (red), while dashed lines indicate the corresponding steady-state operating values. The normalized electricity tariff is shown for reference. The optimized setpoints consistently decrease during high-tariff periods and increase during low-tariff periods, demonstrating stable and reliable tariff-aware control under continuous real-time digital twin operation.

To investigate convergence and structural stability in the optimized flowrate schedules produced by the proposed framework, we analyze the temporal consistency of the predicted 168-hour setpoint trajectories using a rolling-aligned visualization. At each hourly optimization step t , the digital twin generates an optimized flowrate trajectory over the subsequent 168-hour prediction horizon for each manipulated variable (concentrate, permeate, and recirculation). For a given flowrate channel, this trajectory is denoted by $\mathbf{x}_t = [x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(168)}] \in \mathbb{R}^{168}$, where $\mathbf{x}_t$ represents the predicted hourly setpoints starting at time t and extending over the next seven days.

Because the optimization is performed every hour, the first element of $\mathbf{x}_t$ corresponds to different hours within the weekly calendar across successive optimization steps. To enable meaningful comparison, we align all predicted trajectories so that the first data point corresponds to the same hour-of-week in the weekly calendar, which in this study is defined as Monday at 12:00 AM.

Let $\mathbf{x}_t \in \mathbb{R}^{168}$ denote the optimized flowrate trajectory generated at hour t . The aligned trajectory is obtained via a circular shift defined as $\tilde{\mathbf{x}}_t = \text{Roll}(\mathbf{x}_t, -((t-1) \bmod 168))$, where $\text{Roll}(\cdot)$ denotes a circular shift operator. This transformation aligns all trajectories to the same weekly reference, enabling direct comparison across optimization steps under the periodic electricity tariff.

Figure 12 illustrates this alignment procedure for the concentrate flowrate. Figure 12(a) shows the aligned

168-hour trajectory generated at the 4th hour of operation. Because this trajectory begins three hours after the fixed weekly reference, the first three entries are wrapped from the end of the vector and highlighted in red vertical line. Figure 12(b) presents the aligned trajectory generated exactly one week later ($t+168$ hours). Owing to the weekly repetition of the tariff structure, the aligned trajectories in Figures 12(a) and 12(b) exhibit strong similarity, indicating consistency of the optimized response across weekly cycles. The same behavior is observed for another representative hour (10th hour of operation), as shown in Figures 12(c) and 12(d).

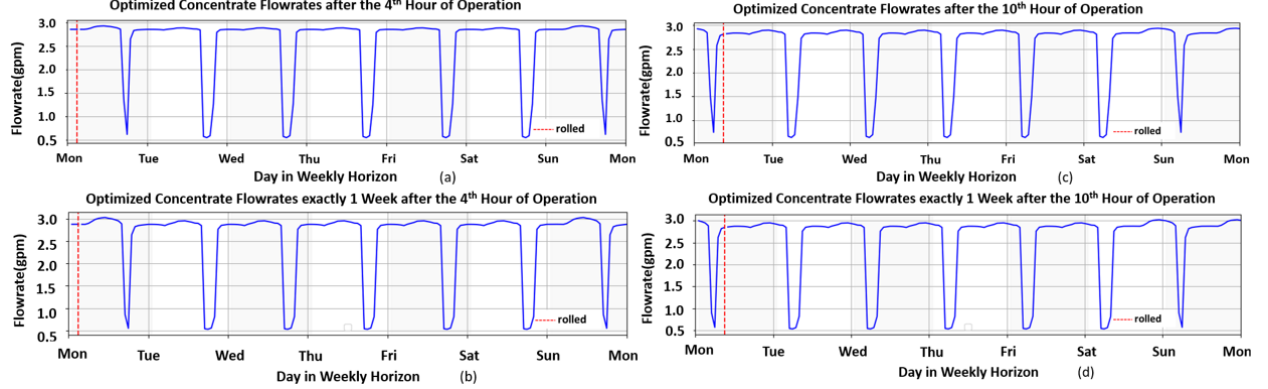


Figure 12: Rolling-aligned optimized concentrate flowrate trajectories (x-axis: Day in Weekly Horizon). (a) Aligned 168-hour trajectory generated at the 4th hour of operation, circularly shifted by three hours relative to the fixed weekly reference (wrapped entries shown in red). (b) Aligned trajectory generated exactly one week after the 4th hour, exhibiting similar structure due to the weekly periodicity of the electricity tariff. (c) Aligned trajectory generated at the 10th hour of operation, circularly shifted by nine hours. (d) Corresponding aligned trajectory generated one week later, showing consistent weekly behavior.

Figure 13 further examines short-term stability by displaying aligned concentrate flowrate trajectories generated at four consecutive optimization steps (5th, 6th, 7th, and 8th hours of operation). Despite limited training data during these early hours, the aligned trajectories exhibit consistent structure and magnitude after circular alignment. This observation demonstrates that the optimization produces stable and repeatable weekly flowrate schedules even during early-stage operation, supporting the robustness of the digital twin under data-sparse conditions.

Next, we quantitatively assess the convergence behavior of the flowrate optimization framework using the aligned setpoint trajectories introduced above. Let $\mathbf{x}^{(t)} \in \mathbb{R}^{168}$ denote the aligned 168-hour setpoint trajectory generated at optimization step t for a given flowrate channel, and let $\mathbf{x}^{(t+k)}$ denote the corresponding trajectory generated k hours later.

We first evaluate long-term (weekly) convergence by comparing trajectories separated by exactly one prediction horizon ($k = 168$). To this end, we define a normalized root-mean-square (RMS) difference metric

$$D_k(t) = \frac{1}{k} \sqrt{\frac{\sum_{i=1}^k \left(x_i^{(t+k)} - x_i^{(t)} \right)^2}{\sum_{i=1}^k \left(x_i^{(t)} \right)^2}},$$

which is dimensionless and quantifies the fractional change in trajectory shape relative to the overall magnitude of the reference trajectory. Values of $D_{168}(t)$ are computed for each optimization step and plotted as a function of time for the concentrate, permeate, and recirculation flowrates in Figures 14(a)–(c). A decreasing trend in $D_{168}(t)$ indicates increasing consistency of the optimized weekly schedules, suggesting convergence of both the model and the associated optimizer under recurring tariff conditions.

To complement this analysis, we also examine short-term (hour-to-hour) convergence by computing the

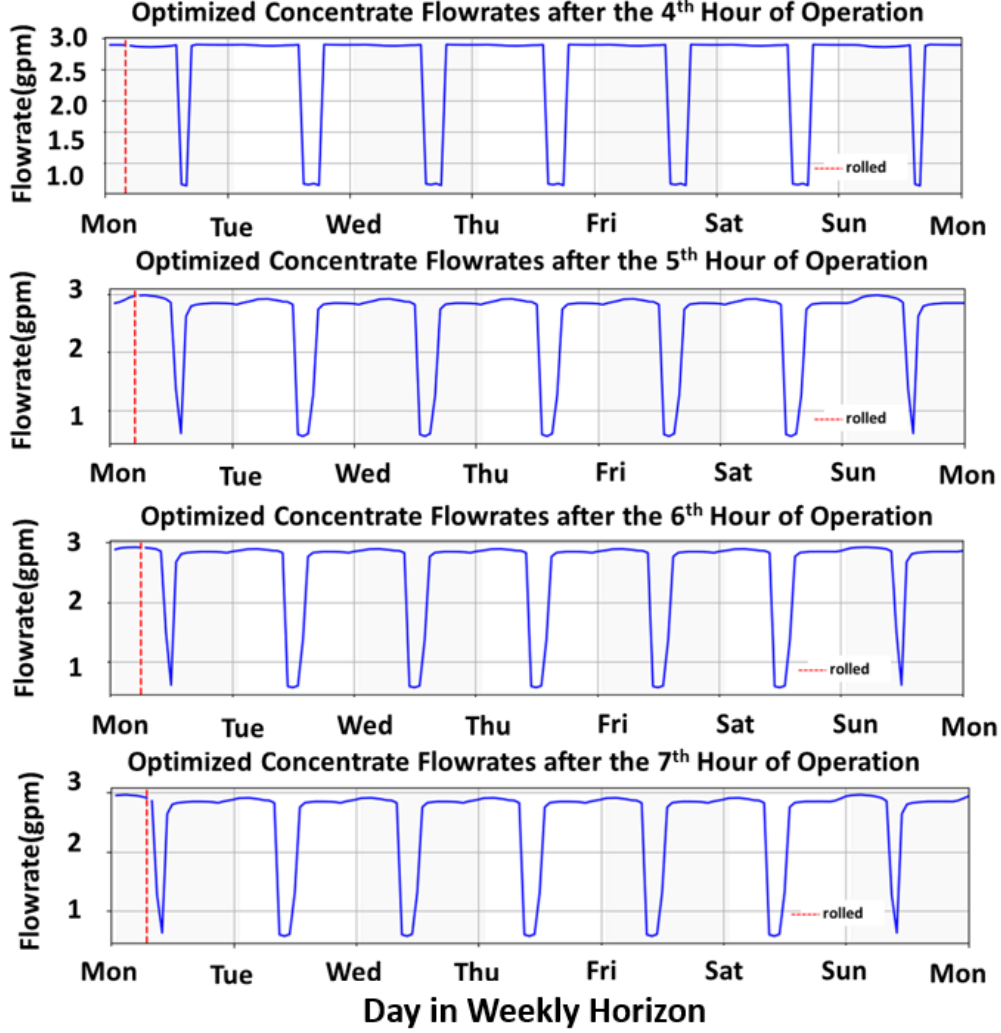


Figure 13: Rolling-aligned optimized concentrate flowrate trajectories for four consecutive optimization steps (5th, 6th, 7th, and 8th hours of operation) (x-axis: Day in Weekly Horizon). Each trajectory is circularly aligned to the fixed weekly reference (Monday 12:00 AM), with wrapped entries indicated by red markers. The strong similarity across consecutive aligned trajectories demonstrates short-term stability of the optimized weekly schedules, even during early-stage operation with limited training data.

same normalized RMS metric for trajectories separated by one hour, defined as

$$D_1(t) = \frac{1}{k} \sqrt{\frac{\sum_{i=1}^k \left(x_i^{(t+1)} - x_i^{(t)} \right)^2}{\sum_{i=1}^k \left(x_i^{(t)} \right)^2}}.$$

Figures 15(a)–(c) show the resulting $D_1(t)$ values for the three flowrate channels. As with the weekly comparison, a clear decreasing trend is observed, indicating progressive stabilization of the optimized setpoint trajectories across successive hourly updates.

Taken together, the results in Figures 14 and 15 demonstrate convergence of the optimized flowrate schedules at both weekly and hourly time scales. This multi-scale convergence behavior confirms that the proposed digital twin framework yields stable, repeatable control policies as training data accumulate and the model matures. These results support the feasibility of extending the proposed framework to full scale water reuse facilities, where similar tariff aware operation could yield substantial operational cost savings.

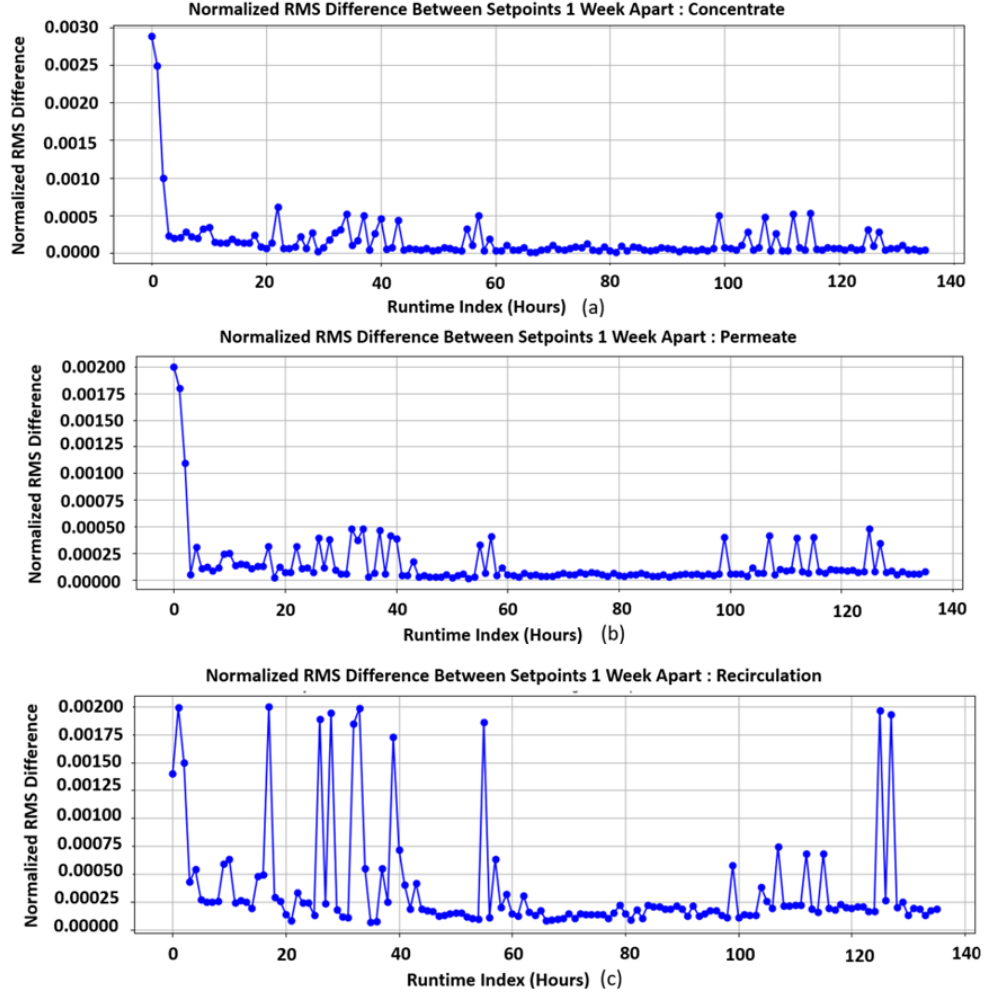


Figure 14: Normalized RMS difference between aligned setpoint trajectories separated by one week ($k = 168$) for (a) concentrate, (b) permeate, and (c) recirculation flowrates.

The two-week deployment demonstrates that the proposed digital twin framework achieves stable, interpretable, and economically consistent operation under dynamic electricity tariffs. The optimized trajectories exhibit strong weekly periodicity aligned with the tariff structure, indicating that the controller internalizes economic patterns rather than responding myopically to short-term fluctuations. This behavior confirms that the receding horizon successfully translates economic signals into coordinated operational adjustments.

An important open challenge concerns improving computational scalability for long-horizon optimization. The weekly electricity tariff exhibits repeated patterns, with identical tariff values recurring across multiple hours. Under the assumption that the learned model is time-invariant over the prediction horizon, this structural regularity suggests that dimensionality reduction strategies may be possible. Such approaches may become particularly relevant when extending the prediction horizon beyond the current weekly window, when adopting finer temporal resolution for control updates, or when coordinating optimization across multiple treatment units in plant-wide operation. In these settings, the dimensionality of the optimization problem increases substantially. Exploiting the repetitive tariff structure could therefore reduce computational burden and improve scalability. Future research will investigate structured parameterizations of the decision variables that preserve optimality while improving computational efficiency for industrial-scale deployment.

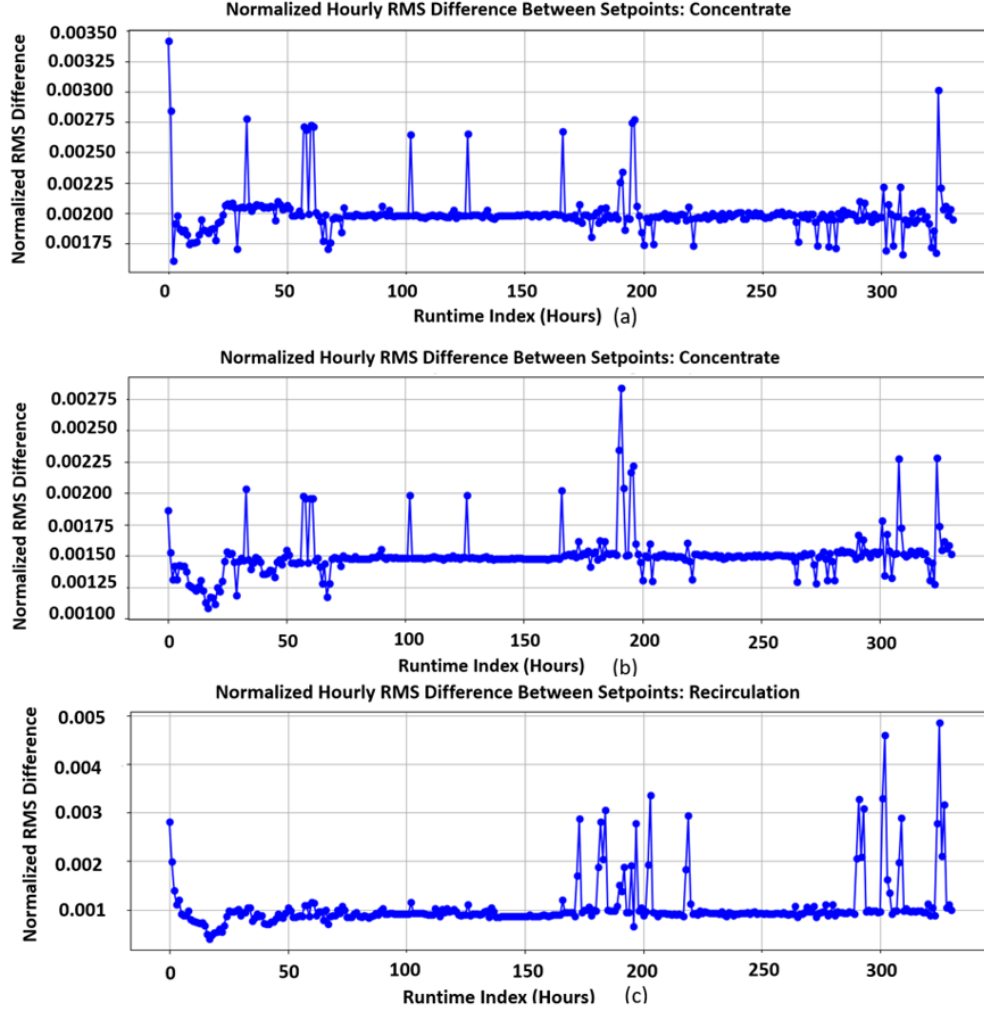


Figure 15: Normalized RMS difference between aligned setpoint trajectories separated by one hour ($k = 1$) for (a) concentrate, (b) permeate, and (c) recirculation flowrates.

4 Conclusion and Future Work

4.1 Conclusion

This work demonstrated a fully automated digital twin framework for online energy cost optimization in a water reuse system. The empirical model-based optimization (EMBO) pipeline integrates continuous model calibration and cost-aware optimization within a closed-loop architecture. Over two weeks of uninterrupted deployment, the framework maintained stable surrogate structure and achieved a 38.33% reduction in operational cost relative to nominal steady-state operation, while exhibiting smooth and feasible control updates. Beyond its demonstrated economic performance, the framework provides interpretability through sparse regression, low computational overhead suitable for hourly operation, and resilience to limited or evolving datasets. These properties are essential for scalable implementation in full-scale treatment facilities. To the best of our knowledge, this study represents one of the first sustained demonstrations of a digital twin based on empirical modeling operating in closed-loop control with a physical water treatment system. These results validate the robustness and reliability of the digital twin in a real pilot environment.

4.2 Future Work

Future extensions will focus on multi-objective optimization that jointly balances energy consumption, membrane fouling, and water quality constraints. Additional research will explore hybrid digital twins that integrate empirical models with physics-based simulators to enhance interpretability and generalization. Integration of adaptive tariff forecasting, anomaly detection, and large-scale supervisory control will further advance the framework toward industrial ready, autonomous operation in sustainable water reuse systems.

Acknowledgement(s)

This material is based upon work supported by the National Alliance for Water Innovation (NAWI), funded by the U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy (EERE), Industrial Technologies Office (ITO), under Funding Opportunity Announcement DE-FOA-0001905, specifically project 5.08 with title “Advanced Process Controls: Autonomous Control and Optimization”.

Funding(s)

This material is based upon work supported by the National Alliance for Water Innovation (NAWI), funded by the U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy (EERE), Industrial Technologies Office (ITO), under Funding Opportunity Announcement DE-FOA-0001905, specifically project 5.08 with title “Advanced Process Controls: Autonomous Control and Optimization”.

A Supplementary Information

We provide the explicit linear inequality coefficients defining the convex polytope used to approximate the empirically realizable operating region described in subsection 2.3. Each inequality corresponds to a supporting half-space of the three-dimensional convex hull of feasible setpoints.

Table 4: Linear inequality constraints derived from 3D convex hull of feasible setpoints. Each row corresponds to a facet: $a_{perm}F_{perm} + a_{conc}F_{conc} + a_{rcyc}F_{rcyc} \leq b$.

a_{perm}	a_{conc}	a_{rcyc}	b	Inequality
0.9570	-0.0166	0.2897	4.1146	$\leq$
1.0000	0.0000	0.0000	3.9000	$\leq$
-0.8830	-0.0859	0.4615	-0.2337	$\leq$
0.8034	0.2373	0.5461	5.1699	$\leq$
-0.6034	0.4872	0.6314	1.8571	$\leq$
-0.7255	0.6829	-0.0854	0.3798	$\leq$
-0.8369	0.5325	-0.1268	-0.7608	$\leq$
-0.8453	0.5265	-0.0908	-0.7183	$\leq$
-0.8346	0.4793	-0.2715	-1.0022	$\leq$
0.9954	-0.0376	-0.0885	3.7054	$\leq$
0.9342	-0.2626	0.2417	3.6139	$\leq$
-0.0097	-0.9959	0.0904	-0.3721	$\leq$
-0.0172	-0.9978	0.0645	-0.4193	$\leq$
0.5358	-0.1478	-0.8313	1.3948	$\leq$
0.2429	-0.3339	-0.9108	0.1518	$\leq$
-0.6059	0.2064	-0.7683	-1.2978	$\leq$
-0.7109	0.3281	-0.6220	-1.2304	$\leq$
-0.1541	-0.1541	-0.9760	-0.7089	$\leq$
-0.5415	0.1692	-0.8235	-1.2116	$\leq$
0.1688	-0.2638	-0.9497	-0.0021	$\leq$
0.2916	-0.1106	-0.9501	0.6173	$\leq$
0.0000	0.0000	-1.0000	0.0000	$\leq$
-0.5774	-0.5774	-0.5774	-1.7321	$\leq$

After 53rd Hour of Operation

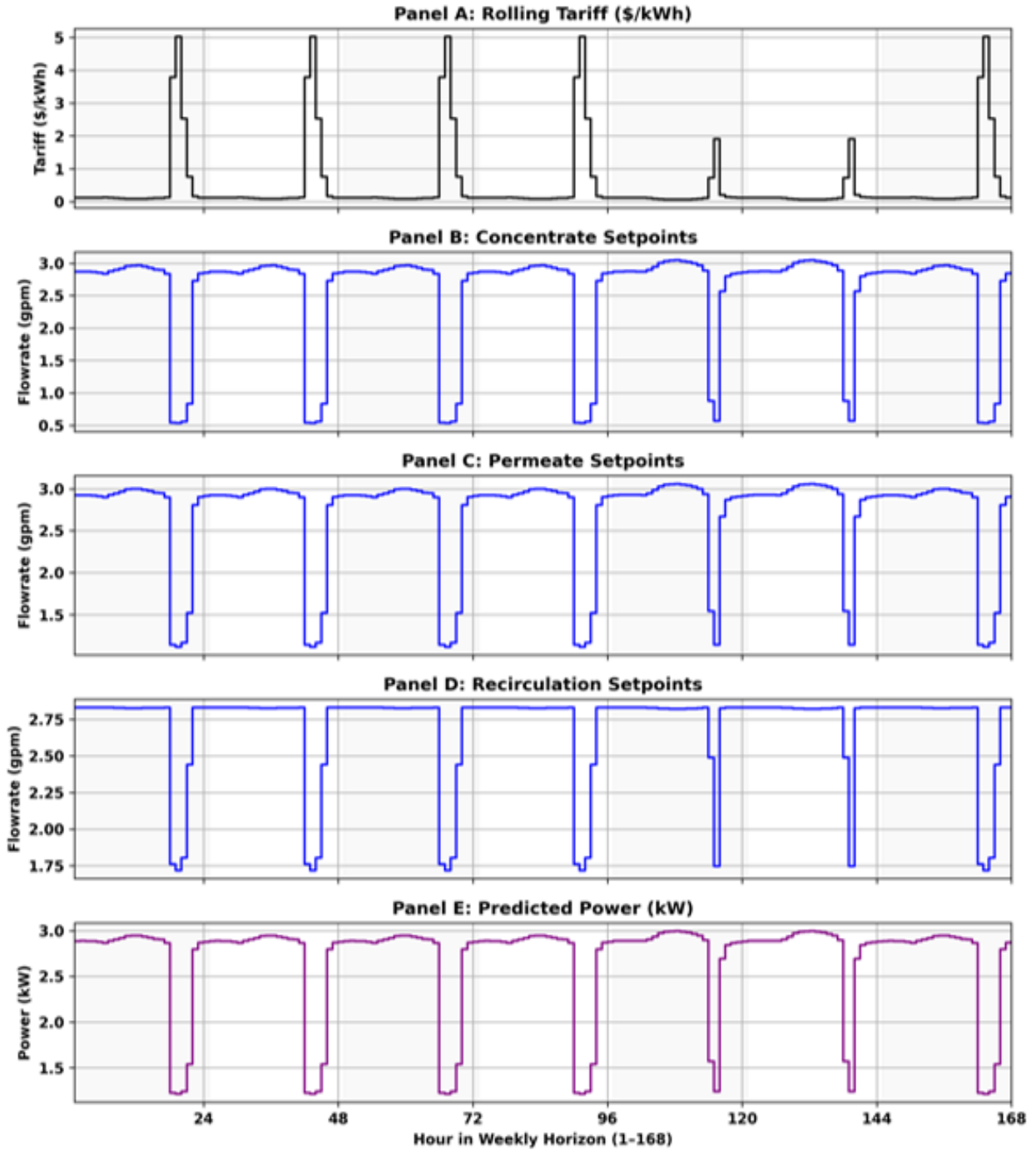


Figure 16: Optimized flowrates, predicted power, and tariff structure at the 53rd hour of operation. The results demonstrate consistent tariff-aware behavior comparable to the representative case discussed in the main text.

After 300th Hour of Operation

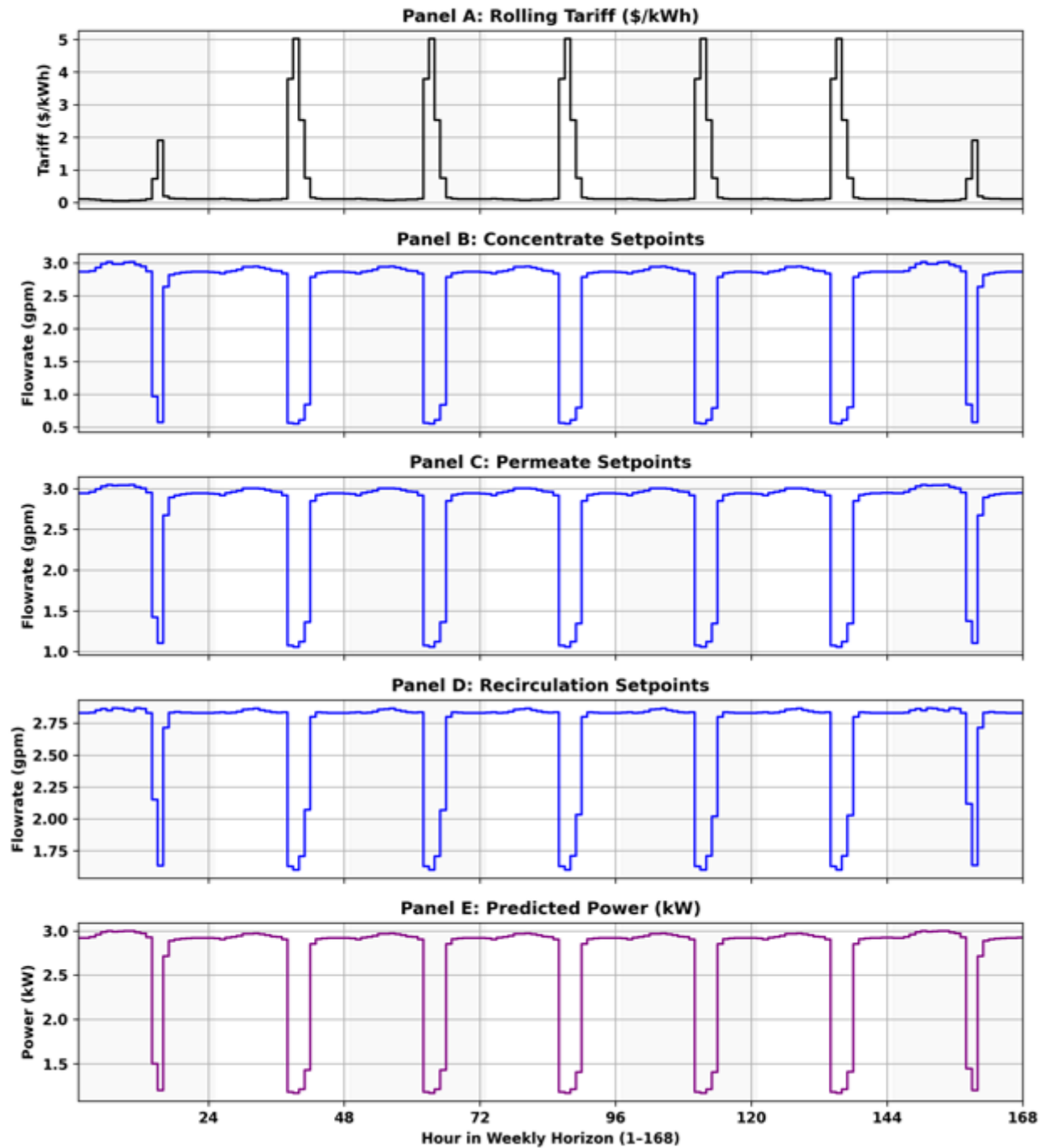


Figure 17: Optimized flowrates, predicted power, and tariff structure at the 300th hour of operation. The optimized schedules exhibit stable and repeatable cost-aware responses under the recurring tariff pattern.

References

- (1) Møller, J. K.; Goranović, G.; Brath, P.; Madsen, H. *Communications Engineering* **2022**, *1*, 23.
- (2) Cuba, Y.; Avila, D.; Quiza, R.; Marichal, G. N. *Water Reuse* **2025**, *15*, 509–527.
- (3) Conejos Fuertes, P.; Martínez Alzamora, F.; Hervás Carot, M.; Alonso Campos, J. *Urban Water Journal* **2020**, *17*, 704–713.
- (4) Glueckstern, P.; Anahory, D. *Desalination and Water Treatment* **2011**, *31*, 172–177.
- (5) Ghorbani Bam, P.; Rezaei, N.; Roubanis, A.; Austin, D.; Austin, E.; Tarroja, B.; Takacs, I.; Villez, K.; Rosso, D. *Water* **2025**, *17*, 2957.
- (6) Ho, M. H.; Hnaien, F.; Dugardin, F. *Computers & Operations Research* **2022**, *144*, 105788.
- (7) Han, Y.; Lam, J. C.; Li, V. O.; Newbery, D.; Guo, P.; Chan, K. *Energy Policy* **2024**, *192*, 114230.
- (8) Gjorgievski, V. Z.; Velkovski, B.; Markovski, B.; Cundeva, S.; Markovska, N. *Energy* **2024**, *313*, 134078.
- (9) Soleimanzade, M. A.; Kumar, A.; Sadrzadeh, M. *Applied Energy* **2022**, *317*, 119184.
- (10) Liu, X.; Wang, J.-S.; Zhang, S.-B.; Guan, X.-Y.; Gao, Y.-Z. *Renewable Energy* **2024**, *237*, 121874.
- (11) Liu, Y.; Feng, J.; Lu, J.; Zhou, S. *Advanced Engineering Informatics* **2024**, *62*, 102592.
- (12) Moazeni, F.; Khazaei, J. *Renewable Energy* **2021**, *167*, 395–408.
- (13) Barata, J.; Kayser, I. *Technovation* **2024**, *134*, 103025.
- (14) Sharma, P.; Bora, B. J.; Deepanraj, B.; Jarin, T. In *2024 1st International Conference on Trends in Engineering Systems and Technologies (ICTEST)*, 2024, pp 1–6.
- (15) Shi, Y.; Xu, J.; Du, W. In *Journal of Physics: Conference Series*, 2019; Vol. 1168, p 022044.
- (16) Chacón Ramírez, E.; Albarrán, J. C.; Cruz Salazar, L. A. In *International workshop on service orientation in Holonic and multi-agent manufacturing*, 2019, pp 352–365.
- (17) Wang, Z.; Song, H.; Watkins, D. W.; Ong, K. G.; Xue, P.; Yang, Q.; Shi, X. *IEEE Communications Magazine* **2015**, *53*, 216–222.
- (18) Singh, M.; Fuenmayor, E.; Hinchy, E. P.; Qiao, Y.; Murray, N.; Devine, D. *Applied System Innovation* **2021**, *4*, 36.
- (19) Kritzinger, W.; Karner, M.; Traar, G.; Henjes, J.; Sihm, W. *Ifac-PapersOnline* **2018**, *51*, 1016–1022.
- (20) Errandonea, I.; Beltrán, S.; Arrizabalaga, S. *Computers in industry* **2020**, *123*, 103316.
- (21) Jones, D.; Snider, C.; Nassehi, A.; Yon, J.; Hicks, B. *CIRP journal of manufacturing science and technology* **2020**, *29*, 36–52.
- (22) Kapteyn, M. G.; Pretorius, J. V.; Willcox, K. E. *Nature Computational Science* **2021**, *1*, 337–347.
- (23) Patriarca, R.; Simone, F.; Di Gravio, G. *Reliability Engineering & System Safety* **2022**, *226*, 108653.
- (24) Fu, A.; Qiao, R.; Wu, Z. *International Journal of Robust and Nonlinear Control* **2024**, *34*, 9638–9653.
- (25) Menapace, A.; Zanfei, A.; Herrera, M.; Brentan, B. *Water* **2024**, *16*, 1835.
- (26) Zhou, S.-W.; Guo, S.-S.; Xu, W.-X.; Du, B.-G.; Liang, J.-Y.; Wang, L.; Li, Y.-B. *Water Resources Management* **2024**, *38*, 2773–2789.
- (27) Mukherjee, S. EMBO Framework: Empirical Model-Based Optimization for Digital–Physical Twin Control, Accessed: January 2026, 2025.
- (28) Mukherjee, S.; Ghorbani Bam, P.; Ghanem, S.; Melin, A. M.; Tarroja, B. J.; Rosso, D.; Villez, K. In *Proceedings of the IWA Leading Edge Conference on Water and Wastewater Technologies*; Accepted for publication, Houston, TX, USA, 2026.
- (29) Tibshirani, R. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **1996**, *58*, 267–288.
- (30) Tibshirani, R. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **2011**, *73*, 273–282.

- (31) Vidaurre, D.; Bielza, C.; Larranaga, P. *International Statistical Review* **2013**, *81*, 361–387.
- (32) Ostertagová, E. *Procedia engineering* **2012**, *48*, 500–506.
- (33) James, G.; Witten, D.; Hastie, T.; Tibshirani, R.; Taylor, J. In *An Introduction to Statistical Learning: with Applications in Python*; Springer: 2023, pp 229–288.
- (34) Bendel, R. B.; Afifi, A. A. *Journal of the American Statistical association* **1977**, *72*, 46–53.
- (35) Hastie, T.; Qian, J. *Retrieved June* **2014**, *9*, 1–30.
- (36) Yang, L.; Qu, Q.; Hao, Z.; Sha, K.; Li, Z.; Li, S. *Journal of Heredity* **2022**, *113*, 472–478.
- (37) Mukherjee, S. *arXiv preprint arXiv:2501.06814* **2025**.
- (38) Jolly, K., *Machine learning with scikit-learn quick start guide: classification, regression, and clustering techniques in Python*; Packt Publishing Ltd: 2018.
- (39) Hao, J.; Ho, T. K. *Journal of educational and behavioral statistics* **2019**, *44*, 348–361.
- (40) Virtanen, P.; Gommers, R.; Burovski, E.; Oliphant, T. E.; Cournapeau, D.; Weckesser, W.; Peterson, P.; Van Der Walt, S.; Mayorov, N.; Wilson, J., et al. *Zenodo* **2018**.
- (41) Samuelsson, C. In *2015 Federated Conference on Computer Science and Information Systems (FedC-SIS)*, 2015, pp 573–578.
- (42) Ypma, J.; Borchers, H. W.; Eddelbuettel, D.; Ypma, M. J. *R package version* **2018**, *1*.
- (43) Ocampo-Martinez, C., *Model predictive control of wastewater systems*; Springer Science & Business Media: 2010.